

Experiment/Case Study No : 3

Aim:

Implement control structures and form validation; create a grading system based on user-entered marks.

Software/Tools Required

- Visual Studio Code
- Google Chrome (Web Browser)

Experiment Program Code

1. prac3.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Student Grading System</title>

<style>
*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:Arial,Helvetica,sans-serif;
}
body{
    display:flex;
    justify-content:center;
    align-items:center;
    min-height:100vh;
    background:linear-gradient(135deg,#4facfe,#00c6ff);
```

```
padding:20px;
}
.container{
width:430px;
background:#fff;
padding:30px;
border-radius:18px;
box-shadow:0 12px 30px rgba(0,0,0,.2);
}
h2{
text-align:center;
color:#222;
margin-bottom:8px;
}
.subtitle{
text-align:center;
color:#666;
font-size:15px;
margin-bottom:25px;
}
label{
display:block;
font-weight:bold;
margin-bottom:8px;
color:#444;
}
input{
width:100%;
padding:12px;
border:2px solid #ddd;
border-radius:10px;
```

```
    font-size:16px;
    margin-bottom:20px;
}
input:focus{
    outline:none;
    border-color:#2196f3;
}
button{
    width:100%;
    padding:13px;
    border:none;
    border-radius:10px;
    background:#2196f3;
    color:#fff;
    font-size:16px;
    cursor:pointer;
    transition:.3s;
}
button:hover{
    background:#1976d2;
}
.grade-info{
    margin-top:25px;
}
.grade-info h3{
    text-align:center;
    color:#333;
    margin-bottom:10px;
}
table{
    width:100%;
```

```
border-collapse:collapse;
}
table td{
padding:8px;
border-bottom:1px solid #eee;
}
table tr:last-child td{
border-bottom:none;
}
table td:last-child{
text-align:right;
}
#result{
margin-top:20px;
padding:15px;
border-radius:10px;
text-align:center;
font-size:18px;
font-weight:bold;
display:none;
}
.credit{
position:fixed;
bottom:15px;
right:15px;
color:white;
font-size:15px;
font-weight:bold;
text-shadow:1px 1px 4px rgba(0,0,0,.5);
}
</style>
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
<h2> 🎓 Student Grading System</h2>
```

```
<p class="subtitle">
```

Enter your marks to calculate your grade instantly.

```
</p>
```

```
<form id="gradeForm">
```

```
<label for="marks">Enter Marks (0 - 100)</label>
```

```
<input type="number" id="marks" placeholder="Example: 85">
```

```
<button type="submit">Calculate Grade</button>
```

```
</form>
```

```
<div class="grade-info">
```

```
<h3>Grade Scale</h3>
```

```
<table>
```

```
<tr><td>A+</td><td>90 - 100</td></tr>
```

```
<tr><td>A</td><td>80 - 89</td></tr>
```

```
<tr><td>B</td><td>70 - 79</td></tr>
```

```
<tr><td>C</td><td>60 - 69</td></tr>
<tr><td>D</td><td>50 - 59</td></tr>
<tr><td>Fail</td><td>Below 50</td></tr>
</table>
```

```
</div>
```

```
<div id="result"></div>
```

```
</div>
```

```
<div class="credit">
```

Designed by M Hussain Khan | PRN : 25070521505

```
</div>
```

```
<script>
```

```
document.getElementById("gradeForm").addEventListener("submit",function(e){
```

```
e.preventDefault();
```

```
let marks=document.getElementById("marks").value;
```

```
let result=document.getElementById("result");
```

```
if(marks==""){
```

```
result.style.display="block";
```

```
result.style.background="#ffebee";
```

```
result.style.color="#d32f2f";
```

```
result.innerHTML="Please enter your marks.";
```

```
return;
```

```
}
```

```
marks=Number(marks);
```

```
if(marks<0 || marks>100){  
    result.style.display="block";  
    result.style.background="#ffebee";  
    result.style.color="#d32f2f";  
    result.innerHTML="Marks must be between 0 and 100.";  
    return;  
}
```

```
let grade="";  
let performance="";
```

```
if(marks>=90){  
    grade="A+";  
    performance="Outstanding";  
}  
else if(marks>=80){  
    grade="A";  
    performance="Excellent";  
}  
else if(marks>=70){  
    grade="B";  
    performance="Very Good";  
}  
else if(marks>=60){  
    grade="C";  
    performance="Good";  
}  
else if(marks>=50){  
    grade="D";
```

```
performance="Pass";
}
else{
grade="Fail";
performance="Needs Improvement";
}

result.style.display="block";

if(grade=="Fail"){
result.style.background="#ffebee";
result.style.color="#c62828";
}
else{
result.style.background="#e8f5e9";
result.style.color="#2e7d32";
}

result.innerHTML=
"<h3>Result</h3><br>" +
"📊 <b>Marks:</b> " +marks+ " / 100<br><br>" +
"🏆 <b>Grade:</b> " +grade+ "<br><br>" +
"⭐ <b>Performance:</b> " +performance;

});
</script>

</body>
</html>
```


Output

Student Grading System

Ask Gemini

C:/Users/MohammadHussain%20Khan/Downloads/prac3.html

Student Grading System

Enter your marks to calculate your grade instantly.

Enter Marks (0 - 100)

49

Calculate Grade

Grade Scale

A+	90 - 100
A	80 - 89
B	70 - 79
C	60 - 69
D	50 - 59
Fail	Below 50

Result

Marks: 49 / 100

Grade: Fail

Performance: Needs Improvement

Designed by M Hussain Khan | PRN : 25070521505

Student Grading System

Ask Gemini

C:/Users/MohammadHussain%20Khan/Downloads/prac3.html

Student Grading System

Enter your marks to calculate your grade instantly.

Enter Marks (0 - 100)

91

Calculate Grade

Grade Scale

A+	90 - 100
A	80 - 89
B	70 - 79
C	60 - 69
D	50 - 59
Fail	Below 50

Result

Marks: 91 / 100

Grade: A+

Performance: Outstanding

Designed by M Hussain Khan | PRN : 25070521505

Case Study Title

Grading system with email and password validation

Case Study Program Code

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Student Grading System - Secure Login</title>

<style>

*{margin:0;padding:0;box-sizing:border-box;font-family:Arial,Helvetica,sans-serif;}

body{display:flex;justify-content:center;align-items:center;min-
height:100vh;background:linear-gradient(135deg,#4facfe,#00c6ff);padding:20px;}

.container{width:100%;max-width:500px;background:#fff;padding:30px;border-
radius:18px;box-shadow:0 12px 30px rgba(0,0,0,.2);}

h2{text-align:center;color:#222;margin-bottom:8px;}

.subtitle{text-align:center;color:#666;font-size:15px;margin-bottom:25px;}

label{display:block;font-weight:bold;margin-bottom:8px;color:#444;}

input{width:100%;padding:12px;border:2px solid #ddd;border-radius:10px;font-
size:16px;margin-bottom:10px;transition:border-color 0.3s;}

input:focus{outline:none;border-color:#2196f3;}

input.error{border-color:#f44336;}

input.success{border-color:#4caf50;}

.error-msg{color:#f44336;font-size:13px;margin-top:-5px;margin-
```

```
bottom:15px;display:none;}

.btn-group{display:flex;gap:10px;margin-top:10px;}

button{flex:1;padding:13px;border:none;border-
radius:10px;background:#2196f3;color:#fff;font-
size:16px;cursor:pointer;transition:background 0.3s;}

button:hover{background:#1976d2;}

button.secondary{background:#607d8b;}

button.secondary:hover{background:#455a64;}

button.danger{background:#f44336;flex:0 0 50px;}

button.danger:hover{background:#d32f2f;}

.subject-row{display:flex;gap:10px;margin-bottom:10px;}

.subject-row input{margin-bottom:0;}

.grade-info{margin-top:25px;}

.grade-info h3{text-align:center;color:#333;margin-bottom:10px;}

table{width:100%;border-collapse:collapse;}

table td{padding:8px;border-bottom:1px solid #eee;}

table tr:last-child td{border-bottom:none;}

table td:last-child{text-align:right;}

#result{margin-top:20px;padding:15px;border-radius:10px;text-align:center;font-
size:18px;font-weight:bold;display:none;}

.credit{position:fixed;bottom:15px;right:15px;color:white;font-size:15px;font-
weight:bold;text-shadow:1px 1px 4px rgba(0,0,0,.5);}

.hidden{display:none;}

</style>

</head>

<body>
```

```
<div class="container" id="loginContainer">

<h2>🔒 Secure Login</h2>

<p class="subtitle">Enter valid credentials to access the system</p>

<form id="loginForm">

<label for="email">Email Address</label>

<input type="text" id="email" placeholder="example@domain.com">

<div id="emailError" class="error-msg">Invalid email format.</div>

<label for="password">Password</label>

<input type="password" id="password" placeholder="Min 8 chars, 1 Upper, 1 Lower, 1
Number, 1 Special">

<div id="passwordError" class="error-msg">Password must be 8+ chars with upper, lower,
number, and special char.</div>

<button type="submit" style="margin-top:15px;">Login</button>

</form>

</div>
```

```
<div class="container hidden" id="gradeContainer">

<h2>🎓 Symbiosis Institute of technology</h2>

<p class="subtitle">Add subjects and marks to calculate final grade</p>

<div id="subjectsContainer"></div>

<div class="btn-group">

<button type="button" class="secondary" onclick="addSubject()">+ Add Subject</button>

<button type="button" onclick="calculateFinalGrade()">Calculate Grade</button>
```

</div>

<div class="grade-info">

<h3>Grade Scale</h3>

<table>

<tr><td>A+</td><td>90% - 100%</td></tr>

<tr><td>A</td><td>80% - 89%</td></tr>

<tr><td>B</td><td>70% - 79%</td></tr>

<tr><td>C</td><td>60% - 69%</td></tr>

<tr><td>D</td><td>50% - 59%</td></tr>

<tr><td>Fail</td><td>Below 50%</td></tr>

</table>

</div>

<div id="result"></div>

</div>

<div class="credit">Designed by M Hussain Khan | PRN : 25070521505</div>

<script>

const loginForm = document.getElementById("loginForm");

const emailInput = document.getElementById("email");

const passwordInput = document.getElementById("password");

const emailError = document.getElementById("emailError");

const passwordError = document.getElementById("passwordError");

const loginContainer = document.getElementById("loginContainer");

const gradeContainer = document.getElementById("gradeContainer");

```
function validateEmail(email) {  
  
const re = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;  
  
return re.test(String(email).toLowerCase());  
  
}
```

```
function validatePassword(pwd) {  
  
const minLength = /.{8,}/;  
  
const hasUpper = /[A-Z]/;  
  
const hasLower = /[a-z]/;  
  
const hasNumber = /[0-9]/;  
  
const hasSpecial = /[!@#$%^&*(),.?:{}|<>]/;  
  
return minLength.test(pwd) && hasUpper.test(pwd) && hasLower.test(pwd) &&  
hasNumber.test(pwd) && hasSpecial.test(pwd);  
  
}
```

```
loginForm.addEventListener("submit", function(e) {  
  
e.preventDefault();  
  
let isValid = true;
```

```
const email = emailInput.value.trim();  
  
const password = passwordInput.value;
```

```
if (!validateEmail(email)) {  
  
emailError.style.display = "block";  
  
emailInput.classList.add("error");  
  
emailInput.classList.remove("success");  
  
isValid = false;
```



```
} else {  
  
emailError.style.display = "none";  
emailInput.classList.remove("error");  
emailInput.classList.add("success");  
  
}  
  
if (!validatePassword(password)) {  
passwordError.style.display = "block";  
passwordInput.classList.add("error");  
passwordInput.classList.remove("success");  
isValid = false;  
} else {  
passwordError.style.display = "none";  
passwordInput.classList.remove("error");  
passwordInput.classList.add("success");  
}  
  
if (isValid) {  
loginContainer.classList.add("hidden");  
gradeContainer.classList.remove("hidden");  
addSubject();  
}  
});
```

```
let subjectCount = 0;
```

```
function addSubject() {
```

```

subjectCount++;

const container = document.getElementById('subjectsContainer');

const div = document.createElement('div');

div.className = 'subject-row';

div.id = `subject-${subjectCount}`;

div.innerHTML = `

<input type="text" placeholder="Subject Name" class="sub-name" required>

<input type="number" placeholder="Marks" class="sub-marks" min="0" max="100"
required>

<button type="button" class="danger"
onclick="removeSubject(${subjectCount})">x</button>

`;

container.appendChild(div);
}

```

```

function removeSubject(id) {

const el = document.getElementById(`subject-${id}`);

if(el) el.remove();

}

```

```

function calculateFinalGrade() {

const names = document.querySelectorAll('.sub-name');

const marksInputs = document.querySelectorAll('.sub-marks');

const resultDiv = document.getElementById('result');

let totalMarks = 0;

let count = 0;

let valid = true;

```

```
let subjectDetails = [];
```

```
names.forEach((input, index) => {
```

```
const name = input.value.trim();
```

```
const marks = parseFloat(marksInputs[index].value);
```

```
if(name === "" || isNaN(marks)) { valid = false; return; }
```

```
if(marks < 0 || marks > 100) { valid = false; return; }
```

```
totalMarks += marks;
```

```
count++;
```

```
subjectDetails.push({name, marks});
```

```
});
```

```
if(!valid || count === 0) {
```

```
resultDiv.style.display = "block";
```

```
resultDiv.style.background = "#ffebee";
```

```
resultDiv.style.color = "#d32f2f";
```

```
resultDiv.innerHTML = "Please fill all subject names and valid marks (0-100).";
```

```
return;
```

```
}
```

```
const average = totalMarks / count;
```

```
let grade = "";
```

```
let performance = "";
```

```
if(average >= 90){grade="A+";performance="Outstanding";}
```

```
else if(average >= 80){grade="A";performance="Excellent";}
```

```
else if(average >= 70){grade="B";performance="Very Good";}
else if(average >= 60){grade="C";performance="Good";}
else if(average >= 50){grade="D";performance="Pass";}
else{grade="Fail";performance="Needs Improvement";}
```

```
let html = `

### 


```

```
resultDiv.style.display = "block";

if(grade === "Fail"){

resultDiv.style.background = "#ffebee";

resultDiv.style.color = "#c62828";

}else{

resultDiv.style.background = "#e8f5e9";

resultDiv.style.color = "#2e7d32";

}

resultDiv.innerHTML = html;

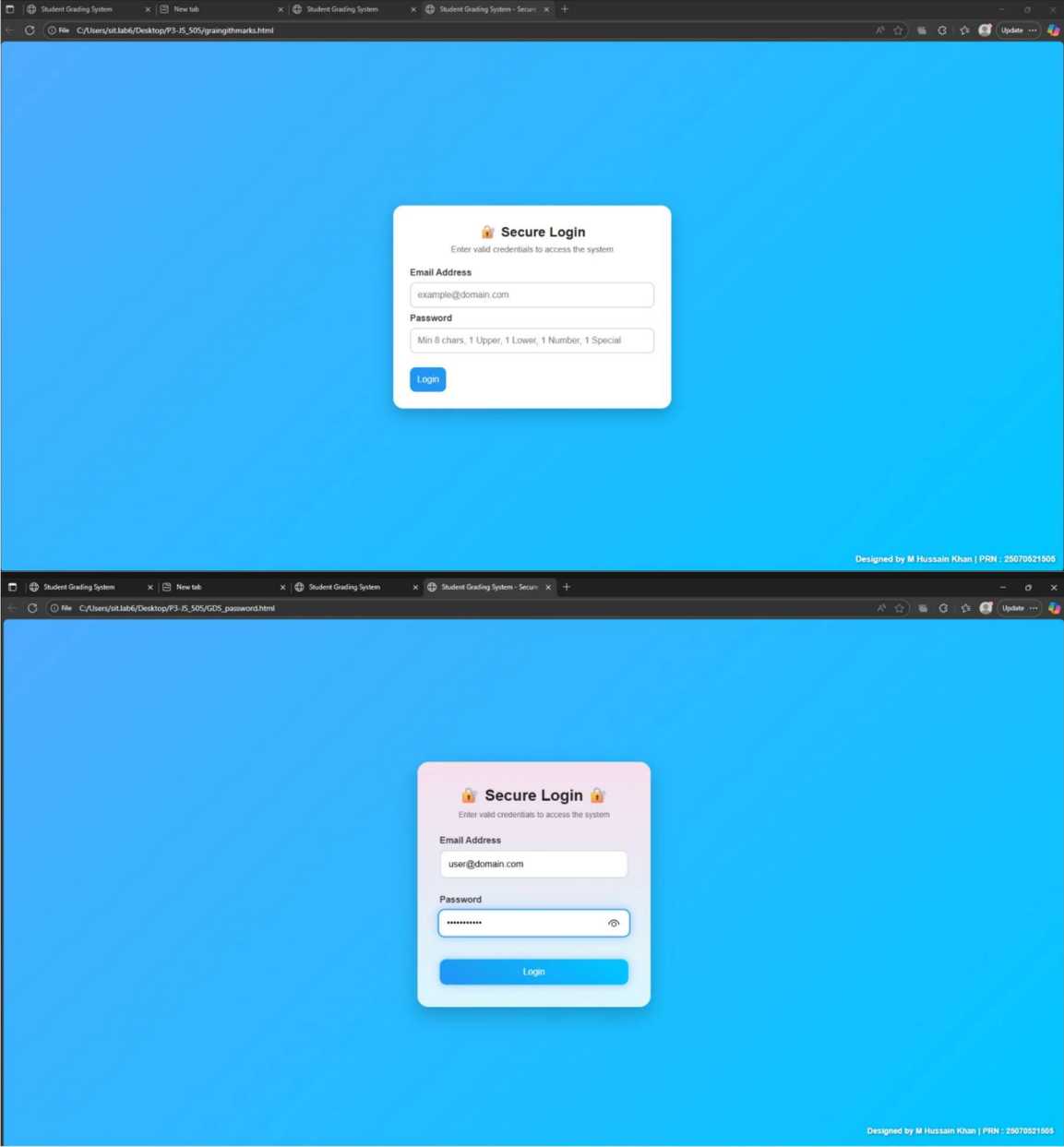
}

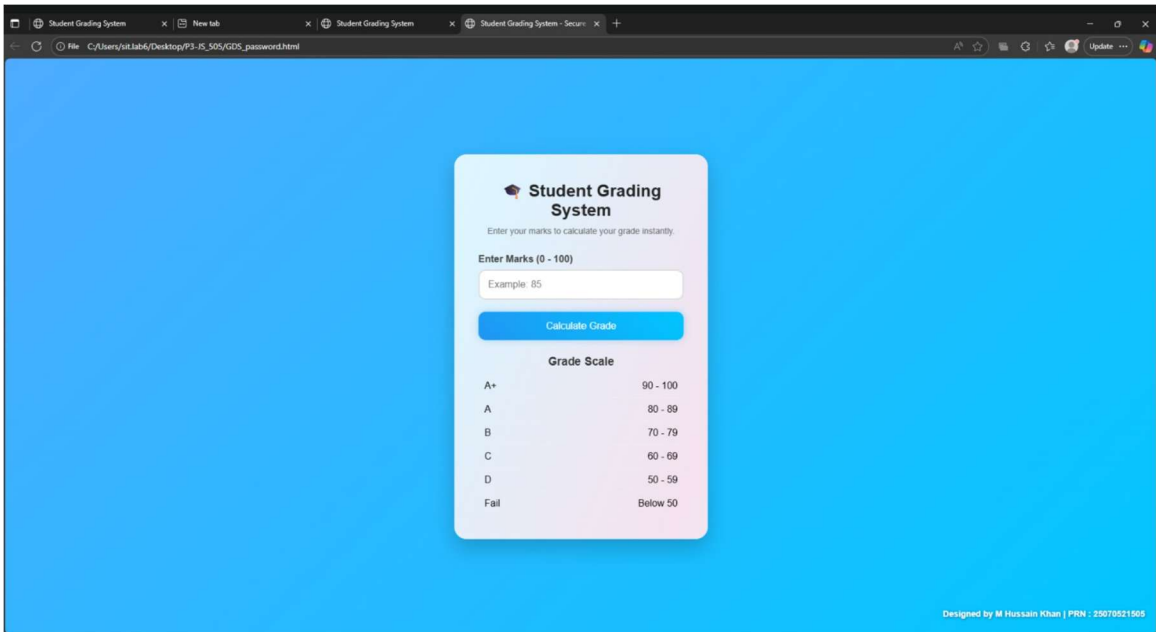
</script>
```

</body>

</html>

Output





Result/Conclusion

The program was executed successfully. User input was validated using form validation, and control structures were used to calculate and display the correct grade and performance based on the marks entered by the user.